Class 26: Calculating Electronic contribution to specific heat

Two classes ago we obtained the following expressions for the total number of electrons $N(E)$ below an energy level E, and for the density of available states $D(E)$

$$N(E) = \frac{1}{6}\pi\left(\frac{2ma^2}{\hbar^2\pi^2}\right)^{\frac{3}{2}} E^{\frac{3}{2}}$$

$$D(E) = \frac{1}{4}\pi\left(\frac{2ma^2}{\hbar^2\pi^2}\right)^{\frac{3}{2}} E^{\frac{1}{2}}$$

Since 'a' represents the extent of the system for nearly free electrons, a^3 represents the volume V of the system.

Therefore we can rewrite the above equations as:

$$N(E) = \frac{V}{6}\pi\left(\frac{2m}{\hbar^2\pi^2}\right)^{\frac{3}{2}} E^{\frac{3}{2}}$$

And

$$D(E) = \frac{V}{4}\pi\left(\frac{2m}{\hbar^2\pi^2}\right)^{\frac{3}{2}} E^{\frac{1}{2}}$$

We have noted earlier that only electrons close to the Fermi energy can gain energy as we raise the temperature. At 0 Kelvin, $f(E) = 1$ for energy levels up to E_f. Therefore,

$$N(E_f) = \frac{V}{6}\pi\left(\frac{2m}{\hbar^2\pi^2}\right)^{\frac{3}{2}} E_f^{\frac{3}{2}}$$

Rearranging, we have:

$$\frac{N(E_f)}{V} = \frac{1}{6}\pi\left(\frac{2m}{\hbar^2\pi^2}\right)^{\frac{3}{2}} E_f^{\frac{3}{2}}$$

The above represents the nearly free electrons, up to the Fermi energy, per unit volume. Therefore it represents the number of free electrons per unit volume, which we had previously designated as N_f.

Similarly, the density of available states per unit volume, at the Fermi energy, is given by:

$$\frac{D(E_f)}{V} = \frac{1}{4}\pi\left(\frac{2m}{\hbar^2\pi^2}\right)^{\frac{3}{2}} {E_f}^{\frac{1}{2}}$$

Which we can designate as $D_V(E_f)$.

Therefore we find that

$$\frac{D_V(E_f)}{N_f} = \frac{3}{2}\frac{1}{E_f}$$

Or,

$$D_V(E_f) = \frac{3}{2}\frac{N_f}{E_f}$$

When we raise the temperature of the system from 0 Kelvin to a temperature T, on a per electron basis, the energy provided is k_BT. Therefore, only electrons within k_BT of the Fermi energy can participate in gaining this energy and move to unoccupied states of higher energies.

On a per unit volume basis, the number of such electrons is given by:

$$D_V(E_f) \times k_BT = \frac{3}{2}\frac{N_f}{E_f}k_BT$$

The energy provided to each of those electrons is

$$\frac{3}{2}k_BT$$

The energy possessed by electrons at temperature T = The number of electrons available × Energy per electron.

Therefore:

$$E = \left(\frac{3}{2}\frac{N_f}{E_f}k_BT\right) \times \left(\frac{3}{2}k_BT\right)$$

Or

$$E = \frac{9}{4}\frac{N_f}{E_f}{k_B}^2T^2$$

Therefore,

$$\frac{dE}{dT} = C_V^e = \frac{9}{2}\frac{N_f}{E_f}{k_B}^2 T$$

Since

$$E_f = k_B T_f$$

We have:

$$C_V^e = \frac{9}{2}\frac{N_f {k_B}^2 T}{k_B T_f} = \frac{9}{2} N_f k_B \left(\frac{T}{T_f}\right)$$

The above expression for C_V^e is therefore the prediction of the Drude-Sommerfeld model.

The classical Drude model predicted:

$$C_V^e = \frac{3}{2} N_f k_B$$

The two expressions primarily differ in the term $\frac{T}{T_f}$. With the Fermi temperature T_f, of the order of 10,000 K, and room temperature T of the order of a few 100 K,

$$\frac{T}{T_f} \approx 0.01$$

The Drude-Sommerfeld model therefore effectively corrects a major shortcoming of the classical Drude model, and hence represents a significant improvement in our efforts to build a model for the properties of solids.

While it is indeed an improvement, the Drude-Sommerfeld model is still only a free electron model. There are no features in the model to enable it explain anisotropy in material properties. The parameter N_f, the number of free electrons per unit volume, is the same regardless of direction, therefore directional variation in properties cannot be explained using this parameter.

The aspect of the material that differentiates between various directions in the material, is its crystal structure. We have not accounted for, or incorporated, the crystal structure of the solid in any way in the model for the material.

Thus far we have treated the wavelike behavior of the nearly free electrons independently, and ignored any interactions between these nearly free electrons and the periodically arranged ionic cores. Even then we have made significant progress in modeling materials, making predictions of material properties, and correcting predictions in earlier models. However to explain anisotropy, we need to understand the interaction between the ionic cores and the wavelike behavior of the nearly free electrons. This is shown in the schematic in Figure 26.1 below.

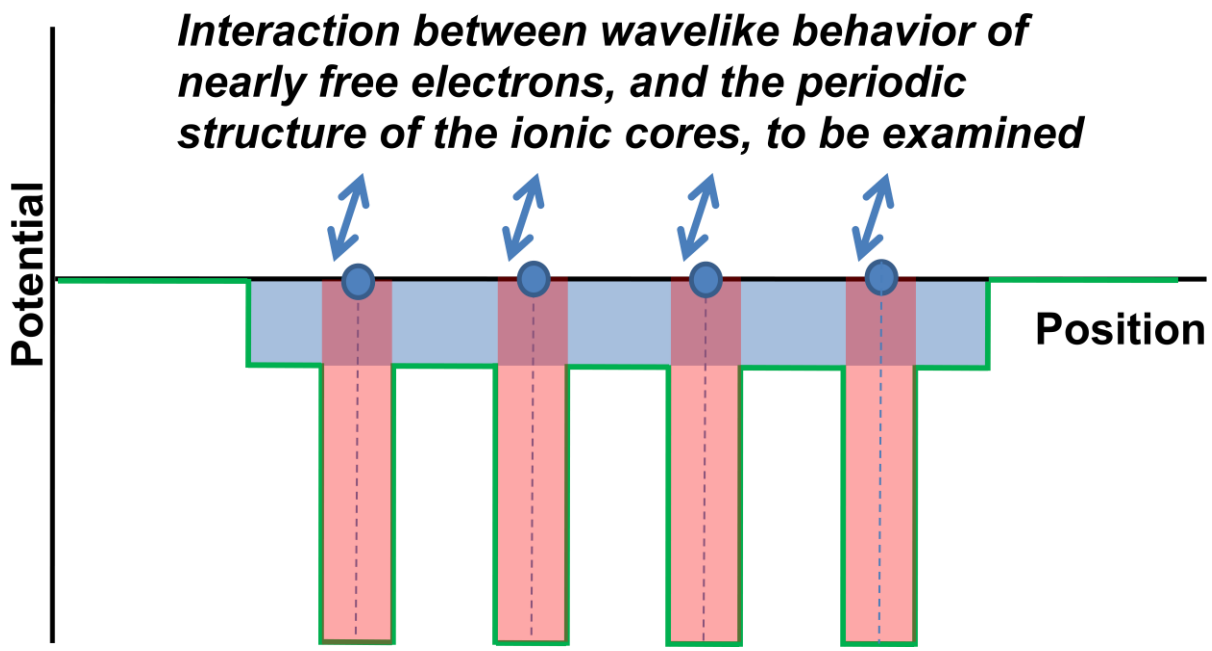


Figure 26.1: A schematic which highlights the interaction between the wavelike behavior of nearly free electrons, and the periodic structure of the ionic cores, that we must now address to improve our model for materials further.

To do the above, we will take a two step process.

1) We will examine the interaction of waves in general with the periodic structure of the ionic cores, which is the diffraction process.
2) We will then take into account the fact that the nearly free electrons in the material are also showing wavelike behavior, and since the diffraction process does not specify the origin of the waves, it will be possible to examine the interaction between the wavelike behavior of the nearly free electrons and the ionic cores, using the same diffraction phenomenon.

However, we note that the wave vector k is in reciprocal length units, or is a vector in 'Reciprocal' space. We have typically studied crystal structures using 'Real' space, which contains distances in ordinary length dimensions and units. To study the interactions between waves and the crystal structures, it helps if they are both in the same kind of space and have the same dimensions. In this context, reciprocal space is seen to capture nuances of the diffraction process much more elegantly and effectively than real space. Therefore, as a first step we have to understand what is reciprocal space, how is it defined, how are crystal structures represented in reciprocal space, and how diffraction is described in reciprocal space. This will be the subject of our discussion in the next few classes. After we have familiarized ourselves with these concepts, we will return to our problem of understanding how waves of nearly free electrons interact with the periodic structure of the material, and what is the consequence of the interaction on material properties. Presumably this will result in the model predicting the presence of anisotropy in material properties.

But first, let us look at reciprocal space.